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ABSTRACT

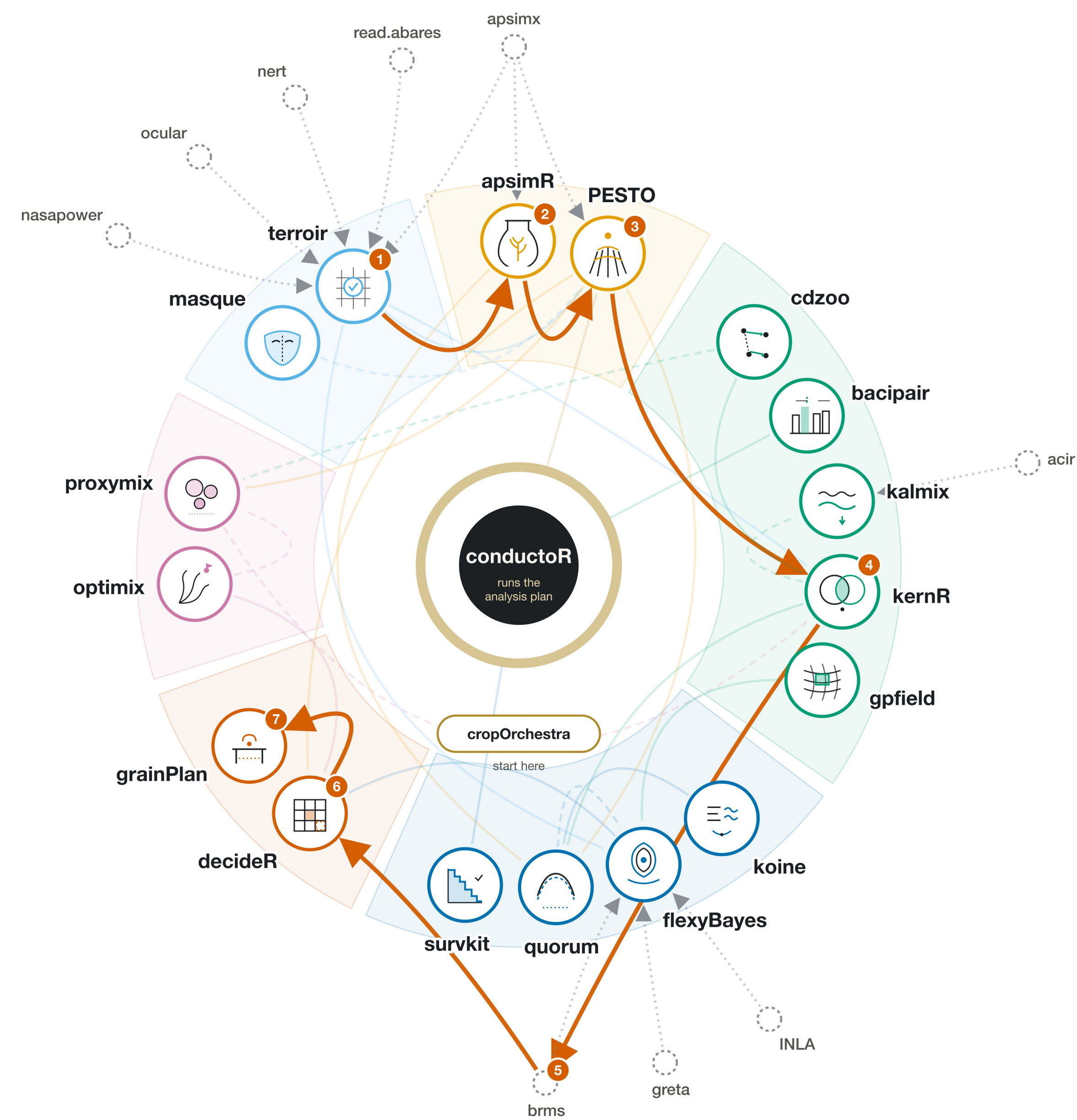
Crop analytics usually joins separate tools by hand, one analysis at a time. The Crop Analytics Orchestra is a set of 20 R packages built to work together. Each package does one job well on its own. All of them return results in one shared format, and each can decline to answer when the data cannot support an answer. The decision packages will not price an answer that was declined earlier. On a synthetic barley belt built on real climate and soil records, the full chain lost 49% less money on nitrogen decisions than an approach that trusts the crop simulator alone.

2 How the packages work together

- One result format.** Every package writes its result in the same structure, defined by the orchestraManifest package. The next package can check the result and use it directly.
- Declining to answer.** A package that cannot answer soundly says so, with its reason, in the same format. Later packages carry that forward instead of guessing.
- conductorR runs the plan.** An analysis plan is a list of steps, each done by one package. conductorR runs the steps in order. Before a result moves to the next step, conductorR checks that it is the right kind of result, and stops the plan if it is not.
- One install.** cropOrchestra lists the packages, installs them and checks that they work together:

```
install.packages("cropOrchestra",  
  repos = c("https://max578.r-universe.dev",  
            "https://cloud.r-project.org"))  
cropOrchestra::cao_install()
```

1 The orchestra



Read the ring clockwise from the top left. Data comes in through terroir (climate and soil records, each with its source) and masque (a shareable synthetic copy of a confidential table). Crop simulation follows: apsimR runs APSIM, and PESTO calibrates it and keeps what remains uncertain. The right side tests and maps: kernR, cdzoo, bacipair, gpfld and kalmix. The bottom holds the statistics: flexyBayes, koine, quorum and survkit. Decisions leave at the lower left through decideR and grainPlan. proxymix and optimix serve every group.

At the centre, conductorR runs a plan across the packages. orchestraManifest is the result format they all write, and cropOrchestra installs the lot. Dotted outer nodes are packages other people maintain. The orange numbers mark the steps of section 3.

Scan the QR code to open this map. Select any package to read what it does and how it connects.

3 One nitrogen decision, told step by step

The data: a synthetic barley belt of 40 sites at real grain-belt coordinates, with real daily climate from NASA POWER and real soil profiles from SoilGrids. Each site is grown for 5 seasons at 6 nitrogen rates from 0 to 150 kg N/ha, with 3 varieties and 3 replicates: 10,800 plot records in all. The record is simulated, so the truth is known exactly: at 12 sites a hidden subsoil constraint caps yield at 57 to 77% of potential and flattens the response above 60 kg N/ha, and the simulator used in the analysis is never told.

The steps, numbered as on the map:

- terroir collects the climate record for every site and keeps its source. Soil profiles come from SoilGrids through apsimx.
- apsimR runs APSIM barley for every site, season, variety and rate. A smaller set of runs trains a fast stand-in for the simulator, an emulator, so that it can be calibrated quickly.
- PESTO calibrates the emulator to the field record at the low rates only, 0 to 60 kg N/ha, and keeps the calibrated parameters as an ensemble of plausible values.
- kernR tests, at the higher rates, whether the field record still agrees with the calibrated model. It rejects the model at 7 of the 12 constrained sites, and wrongly at 2 unconstrained sites.
- brms fits the field record directly with plateau curves, linear and quadratic, weighted by how well each predicts plots held out of the fit. Wherever kernR rejected the model, this field-record fit is used instead.
- decideR prices every candidate rate by expected profit: grain price times yield, less nitrogen cost and a nitrous-oxide shadow price, averaged over the uncertainty. It also prices what one soil test or one strip trial at each site would be worth.
- grainPlan writes one nitrogen rate per site, with the reasons, and keeps every rejection made along the way.

The comparison arm trusts the calibrated simulator everywhere. Both arms are scored by profit foregone against the truth-optimal rate, in AUD/ha, and the whole chain is run again on a re-seeded belt.

4 Results

- 2,960 vs 5,818 AUD/ha** total loss against the planted truth, orchestra chain against the simulator-only approach, 49% less. The same result held on the re-seeded belt (2,549 vs 5,002).
- 1,440 kg N not applied**, 6.2 t CO₂-e/ha avoided, at the constrained sites where extra nitrogen does not pay.
- 164 AUD/ha** the value of one more strip trial at the best site, which shows where the next research dollar pays.

The belt is synthetic: the numbers show the method at work, not what a real paddock would earn.

6 References

Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, 80(1). <https://doi.org/10.18637/jss.v080.i01>

Holzworth, D., Huth, N. I., Fainges, J., Brown, H., Zurcher, E., Cichota, R., Verrall, S., Herrmann, N. I., Zheng, B., & Snow, V. (2018). APSIM Next Generation: Overcoming challenges in modernising a farming systems model. *Environmental Modelling & Software*, 103, 43–51. <https://doi.org/10.1016/j.envsoft.2018.02.002>

Sparks, A. (2018). nasapower: A NASA POWER global meteorology, surface solar energy and climatology data client for R. *Journal of Open Source Software*, 3(30), 1035. <https://doi.org/10.21105/joss.01035>

5 Get it and explore

Open source (MIT). Interactive map, every package page and this study: max578.github.io/orchestra/members (QR code).

